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Semiconductor device and method of fabricating same

Abstract

A semiconductor device includes; a semiconductor substrate including a first region and a second region, a first interlayer insulating layer on the second region, a capping layer disposed on the first interlayer insulating layer, an upper surface of the capping layer includes a first trench, conductive patterns spaced apart on the capping layer, side surfaces of the conductive patterns are aligned with inner side surfaces of the first trench, and a peripheral separation pattern disposed in the first trench to cover the side surfaces of the conductive patterns. The peripheral separation pattern has a first thickness on the side surfaces of the conductive patterns and a second thickness greater than or equal to the first thickness on a lower surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is application is a continuation of U.S. application Ser. No. 18/137,733, filed Apr. 21, 2023, which is a continuation of U.S. application Ser. No. 17/404,757, filed Aug. 17, 2021, now U.S. Pat. No. 11,670,591, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0020108, filed on Feb. 15, 2021, in the Korean Intellectual Property Office, the subject matter of which is hereby incorporated by reference.

BACKGROUND

(1) The inventive concept relates generally to semiconductor devices and methods of fabricating same.

(2) Continuing demands exist for semiconductor devices that are increasingly small in size, diverse and powerful in function(s), and/or inexpensive to make. As a result, contemporary and emerging semiconductor devices are very densely integrated. In order to further increase the integration density of semiconductor devices, it is often necessary to reduce linewidths of patterns and spacing distances between elements. However, highly complex and/or very expensive fabrication technologies are required to reduce pattern linewidths. Accordingly, many studies are ongoing into new approaches, methods and/or technologies that may increase the integration density of semiconductor devices.

SUMMARY

(3) Embodiments of the inventive concept provide semiconductor devices providing improved reliability. Other embodiments of the inventive concept provide methods of fabricating semiconductor devices providing improved manufacturing yield.

(4) According to an embodiment of the inventive concept, a semiconductor device may include; a semiconductor substrate including a first region and a second region, a first interlayer insulating layer covering the second region, a capping layer disposed on the first interlayer insulating layer, wherein an upper surface of the capping layer includes a first trench, conductive patterns spaced apart on the capping layer, wherein side surfaces of the conductive patterns are aligned with inner side surfaces of the first trench, and a peripheral separation pattern disposed in the first trench to cover the side surfaces of the conductive patterns. The peripheral separation pattern has a first thickness on the side surfaces of the conductive patterns and a second thickness greater than or equal to the first thickness on a lower surface.

(5) According to an embodiment of the inventive concept, a semiconductor device may include; a semiconductor substrate including a cell array region and a peripheral region, a bit line on the semiconductor substrate crossing the cell array region, a bit line capping pattern on the bit line, a landing pad placed on the bit line capping pattern and electrically isolated from the bit line, a landing pad separation pattern contacting a side surface of the landing pad, a first interlayer insulating layer covering the peripheral region, a capping layer on the first interlayer insulating layer, wherein an upper surface of the capping layer includes a first trench, conductive patterns spaced apart on the capping layer, wherein side surfaces of the conductive patterns are aligned with inner side surfaces of the first trench, and a peripheral separation pattern filling the first trench to cover the side surfaces of the conductive patterns. The peripheral separation pattern has a first thickness on the side surfaces of the conductive patterns, and a second thickness greater than or equal to the first thickness on a lower surface of the first trench.

(6) According to an embodiment of the inventive concept, a semiconductor device may include; a semiconductor substrate including a cell array region and a peripheral region, a first interlayer

insulating layer covering the peripheral region, a capping layer disposed on the first interlayer insulating layer, wherein an upper surface of the capping layer includes a first trench, conductive patterns spaced apart on the capping layer, wherein side surfaces of the conductive patterns are aligned with inner side surfaces of the first trench, a peripheral separation pattern disposed in the first trench to cover the side surfaces of the conductive patterns, wherein the peripheral separation pattern includes a first peripheral sub-insulating pattern, a second peripheral sub-insulating pattern, and a third peripheral sub-insulating pattern, which are sequentially stacked, landing pads disposed on the cell array region, and a landing pad separation pattern disposed between the landing pads. The landing pad separation pattern includes a first cell sub-insulating pattern, a second cell sub-insulating pattern, and a third cell sub-insulating pattern, which are sequentially stacked, and the second peripheral sub-insulating pattern includes a same material as the second cell sub-insulating pattern.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The making and use of the inventive concept may be more clearly understood upon consideration of the following written description together with the accompanying drawings, in which:
- (2) FIG. 1 is a plan (or top-down) view illustrating a semiconductor device according to embodiments of the inventive concept;
- (3) FIGS. 2A and 2B are respective cross-sectional views illustrating semiconductor devices according to embodiments of the inventive concept taken along lines R1-R1' and R2-R2' of FIG. 1;
- (4) FIGS. 3A and 4A are respective, enlarged cross-sectional views variously illustrating area 'P1' of FIG. 2A, and FIGS. 3B, 3C, 3D, and 4B are respective, enlarged cross-sectional views variously illustrating area 'P2' of FIG. 2A; and
- (5) FIGS. 5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, 5I, 5J, 5K, 5L, 5M, 5N, 5O, 5P and 5Q (hereafter collectively, "FIGS. 5A to 5Q") are related cross-sectional views illustrating methods of fabricating a semiconductor device according to embodiments of the inventive concept.
- (6) The accompanying drawings illustrate general characteristics of semiconductor device(s) and method(s), various structural components and relationships, and/or various materials that may be used in various embodiments of the inventive concept. These drawings are not rendered to scale and may not precisely reflect real-world implementations. For example, for clarity of illustration, thickness of various material layers may be exaggerated.

DETAILED DESCRIPTION

- (7) Throughout the written description and drawings, like reference numbers and labels are used to denote like or similar elements and/or features. Throughout the written description certain geometric terms may be used to highlight relative relationships between elements, components and/or features with respect to certain embodiments of the inventive concept. Those skilled in the art will recognize that such geometric terms are relative in nature, arbitrary in descriptive relationship(s) and/or directed to aspect(s) of the illustrated embodiments. Geometric terms may include, for example: height/width; vertical/horizontal; top/bottom; higher/lower; closer/farther; thicker/thinner; proximate/distant; above/below; under/over; upper/lower; center/side; surrounding; overlay/underlay; etc.
- (8) Figure (FIG. 1 is a plan view illustrating a semiconductor device according to embodiments of the inventive concept, and FIG. 2A is a cross-sectional view further illustrating the semiconductor device of FIG. 1 along lines R1-R1' and R2-R2'.
- (9) Referring to FIGS. 1 and 2A, a semiconductor substrate 301 of the semiconductor device may generally include a cell array region CA and a peripheral region PA. The peripheral region PA may

be disposed substantially around the cell array region CA. Peripheral circuits, such as those commonly used to drive word lines WL and bit lines BL provided in the cell array region CA, may be variously disposed in the peripheral region PA. The peripheral region PA may be alternately be referred to as a core region or a peripheral circuit region.

(10) A device isolation layer **303** may be disposed in the semiconductor substrate **301** to define cell active portions ACTC and a peripheral active portion ACTP. Each of the cell active portions ACTC may be associated with at least one isolation structure. For example, each of the cell active portions ACTC may be a bar-shaped pattern extending in a first direction D1. In some embodiments, the semiconductor substrate **301** may be a single-crystalline silicon substrate or a silicon-on-insulator (SOI) wafer. The device isolation layer **303** may include an oxide liner, a nitride liner, and an insulating gapfill layer. The cell active portions ACTC may be arranged in parallel in the first direction D1, wherein each of the cell active portions ACTC may be arranged such that an end portion thereof is disposed near the center of an adjacent cell active portion ACTC.

(11) The word lines WL may be disposed in such a manner that they cross the cell active portions ACTC. For example, the word lines WL may be disposed in grooves formed in the device isolation layer **303** and the cell active portions ACTC. The word lines WL may extend in a second direction D2 that intersects the first direction D1. In some embodiments, the word lines WL may be buried in the semiconductor substrate **301**.

(12) A first impurity region **305a** may be disposed in each of the cell active portions ACTC between a pair of the word lines WL, and a pair of second impurity regions **305b** may be respectively disposed in opposite edge regions of each of the cell active portions ACTC. The first and second impurity regions **305a** and **305b** may be doped with, for example, one or more N-type impurities. The first impurity region **305a** may correspond to a common drain region, and the second impurity regions **305b** may correspond to a source region.

(13) A first interlayer insulating layer **307** may be disposed on the cell array region CA of the semiconductor substrate **301**. The first interlayer insulating layer **307** may include at least one of silicon oxide, silicon nitride, or silicon oxynitride, and may be a single layer structure or a multi-layered structure.

(14) The bit lines BL may be disposed on the first interlayer insulating layer **307**. The bit lines BL may be disposed in such a manner so as to cross the word lines WL. For example, as illustrated in FIG. 1, the bit lines BL may extend in third direction D3 intersecting both the first direction D1 and the second direction D2. The bit lines BL may include a cell polysilicon pattern **311a**, a cell ohmic pattern **315a**, and a cell metal-containing pattern **317a**, which are sequentially stacked. The cell polysilicon pattern **311a** may include impurity-doped polysilicon. The cell ohmic pattern **315a** may include at least one metal silicide (e.g., cobalt silicide). The cell metal-containing pattern **317a** may include at least one metal (e.g., aluminum, tungsten, and copper). A bit line capping pattern BLC may be disposed on the bit line BL. The bit line capping pattern BLC may include a first bit line capping pattern **319a** and a second bit line capping pattern **331a** stacked on the first bit line capping pattern **319a**. Here, the first bit line capping pattern **319a** and the second bit line capping pattern **331a** may include silicon nitride.

(15) A side surface of the bit line BL and a lower side surface of the bit line capping pattern BLC may be covered with a first bit line spacer **335**. An upper side surface of the bit line capping pattern BLC may be covered with a second bit line spacer **336**. Each of the first and second bit line spacers **335** and **336** may include at least one of silicon nitride, silicon oxide, and silicon oxynitride, and may be a single layer or a multi-layered structure. The second bit line spacer **336** may be provided to have an air gap therein.

(16) The first interlayer insulating layer **307** may be interposed between the bit line BL and the semiconductor substrate **301** and between the first bit line spacer **335** and the semiconductor substrate **301**. A side surface of the first bit line spacer **335** may be aligned to a side surface of the first interlayer insulating layer **307**.

(17) The bit line BL may be electrically connected to the first impurity region **305a** through a bit line contact DC. The bit line contact DC may include impurity-doped polysilicon. A recess region **313** may be formed in an upper portion of the first impurity region **305a** and in an upper portion of the device isolation layer **303** adjacent thereto. The bit line contact DC may be disposed in the recess region **313**. A space between a lower side surface of the bit line contact DC and an inner surface of the recess region **313** may be filled with a gapfill insulating pattern **316**. The gapfill insulating pattern **316** may include at least one of silicon nitride or silicon oxide, and may be a single layer structure or a multi-layered structure.

(18) Storage node contacts BC may be disposed between an adjacent pair of the bit lines BL. The storage node contacts BC may be spaced apart from each other. The storage node contacts BC may include impurity-doped polysilicon. The storage node contacts BC may be island-shaped patterns, which are spaced apart from each other in a plan view. The storage node contacts BC may be provided to penetrate the first interlayer insulating layer **307** and to be in contact with the second impurity regions **305b**.

(19) A storage node ohmic layer **341a** may be disposed on the storage node contact BC. The storage node ohmic layer **341a** may include at least one metal silicide (e.g., cobalt silicide).

(20) A landing pad LP may be disposed on the storage node ohmic layer **341a**. The landing pad LP may include at least one metal (e.g., tungsten, aluminum, and copper). An upper portion of the landing pad LP may cover an upper surface of the bit line capping pattern BLC. A center of the landing pad LP may be shifted (or offset) from a center of the storage node contact BC in the second direction D2. A portion of the bit line BL may vertically overlap the landing pad LP.

(21) Adjacent landing pads LP may be separated by a landing pad separation pattern **347a**. The landing pad separation pattern **347a** may include at least one of silicon nitride, silicon oxide, and silicon carbon nitride, and may be a single layer structure or a multi-layered structure. The landing pad separation pattern **347a** may extend into the second bit line capping pattern **331a** and the second bit line spacer **336**.

(22) An etch stop layer EL may be disposed on the landing pads LP and the landing pad separation pattern **347a**. The etch stop layer EL may include a silicon nitride layer. Bottom electrodes BE may be provided to penetrate the etch stop layer EL and to be in contact with the landing pads LP.

Portions of upper side surfaces of the bottom electrodes BE may contact a supporting pattern SP. Surfaces of the bottom electrodes BE and the supporting pattern SP may be covered with a dielectric layer DL. An upper electrode UE may be disposed on the dielectric layer DL. The bottom electrode BE, the dielectric layer DL, and the upper electrode UE may constitute a capacitor CAP.

(23) A third interlayer insulating layer IL may be stacked on the upper electrode UE. A cell contact CT1 may be provided to penetrate the third interlayer insulating layer IL and to contact the upper electrode UE. A cell interconnection line UM1 may be disposed on the upper electrode UE.

(24) A peripheral gate electrode **323b** may be disposed on the peripheral region PA of the semiconductor substrate **301**. A peripheral gate insulating pattern **309b** may be interposed between the peripheral gate electrode **323b** and the semiconductor substrate **301**. A peripheral gate capping pattern **319b** may be disposed on the peripheral gate electrode **323b**. Peripheral source/drain regions **327** may be disposed in portions of the semiconductor substrate **301** and at both sides of the peripheral gate electrode **323b**. Opposite side surfaces of the peripheral gate electrode **323b** may be covered with peripheral spacers **325**. The peripheral gate electrode **323b** may include a stacked combination of a peripheral polysilicon pattern **311b**, a peripheral ohmic pattern **315b**, and a peripheral metal-containing pattern **317b**. The peripheral gate capping pattern **319b** and the peripheral spacer **325** may be covered with a second interlayer insulating layer **329**. A second capping layer **331** may be disposed on the second interlayer insulating layer **329**.

(25) Peripheral conductive patterns **343bw** may be disposed on the second capping layer **331**. An upper surface of the peripheral conductive patterns **343bw** may be disposed at the same "height" (e.g., a disposition measured with respect to an arbitrarily selected horizontal plane, such as the

upper surface of the semiconductor substrate **301**) as upper surfaces of the landing pads LP. First peripheral contacts **343bc** may be provided to penetrate the second capping layer **331** and the second interlayer insulating layer **329** and may electrically connect the peripheral source/drain regions **327**, respectively. The first peripheral contacts **343bc** and the peripheral conductive patterns **343bw** may be connected to form a single element. A peripheral ohmic layer **341b** may be interposed between the first peripheral contacts **343bc** and the peripheral source/drain regions **327**. A peripheral separation pattern **347b** may be interposed between the peripheral conductive patterns **343bw**. The etch stop layer EL may extend to cover a portion of the peripheral conductive patterns **343bw** and the peripheral separation pattern **347b**. The third interlayer insulating layer IL may cover the etch stop layer EL in the peripheral region PA. Second peripheral contacts MC may be provided to penetrate the third interlayer insulating layer IL and contact the peripheral conductive patterns **343bw**. Peripheral interconnection lines UM2 may be disposed on the third interlayer insulating layer IL and may contact the second peripheral contacts MC.

(26) FIGS. **3A** and **4A** are respective, enlarged cross-sectional views further illustrating area 'P1' of FIG. **2A**, and FIGS. **3B**, **3C**, **3D**, and **4B** are respective, enlarged cross-sectional views further illustrating area 'P2' of FIG. **2A**.

(27) Referring to FIGS. **2A**, **3A**, and **3B**, each of the cell and peripheral polysilicon patterns **311a** and **311b** may have a first thickness T1. In some embodiments, the cell ohmic pattern **315a** and the peripheral ohmic pattern **315b** may include the same material and may have the same thickness. The cell metal-containing pattern **317a** and the peripheral metal-containing pattern **317b** may include the same material and may have the same thickness (e.g., a second thickness T2). A thickness TBL of the bit line BL on the first interlayer insulating layer **307** may be equal to a thickness TGE of the peripheral gate electrode **323b**. The first bit line capping pattern **319a** and the peripheral gate capping pattern **319b** may include the same material and may have the same thickness (e.g., a third thickness T3). The second bit line capping pattern **331a** and the second capping layer **331** may include the same material and may have the same thickness (e.g., a fourth thickness T4). The landing pad LP on the second bit line capping pattern **331a** may have a fifth thickness T5, and the peripheral conductive pattern **343bw** may include the same material as the landing pad LP and may have the fifth thickness T5. A thickness of the bit line capping pattern BLC may be equal to the sum of the third thickness T3 of the peripheral gate capping pattern **319b** and the fourth thickness T4 of the second capping layer **331**.

(28) The peripheral separation pattern **347b** and the landing pad separation pattern **347a** may include the same insulating material. The landing pad separation pattern **347a** may have a first width W1. The landing pad separation pattern **347a** may be provided to fill a space between the landing pads LP. In some embodiments, a void or an air gap may be included in the landing pad separation pattern **347a**.

(29) An upper surface of the second capping layer **331US** may include a first trench **331t**. An inner surface of the first trench **331t** may be aligned with a side surface of the peripheral conductive pattern **343bw**. A lower surface of the first trench **331t** may be spaced apart from the second interlayer insulating layer **329**. Under the lower surface of the first trench **331t**, the second capping layer **331** may have a fourth remaining thickness T4r. Under a lower surface of the landing pad separation pattern **347a**, the second bit line capping pattern **331a** may have the fourth remaining thickness T4r.

(30) Referring to FIGS. **3A** and **3B**, the peripheral separation pattern **347b** may conformally cover the side surface of the peripheral conductive pattern **343bw** and an inner side surface and a lower surface of the first trench **331t**. In this regard, the term "conformally cover" denotes a result in which a position-independent, substantially-constant thickness is achieved by one material layer in relation to at least one other material layer). For example, the peripheral separation pattern **347b** may have a sixth thickness T6 on the side surface of the peripheral conductive pattern **343bw** and may have the sixth thickness T6 on the lower surface of the first trench **331t**. The peripheral

separation pattern **347b** may have a second width **W2** greater than the first width **W1**. The sixth thickness **T6** may be greater than or equal to half of the first width **W1**, and may be less than half of the second width **W2**.

(31) Alternately, referring to FIG. **3C**, the peripheral separation pattern **347b** may have a sixth side thickness **T6s** on the side surface of the peripheral conductive pattern **343bw** and may have a sixth bottom thickness **T6b** on the lower surface of the first trench **331t**. The sixth bottom thickness **T6b** may be greater than the sixth side thickness **T6s**. An upper surface of the peripheral separation pattern **347b** may have a peripheral recess region **347r**.

(32) An upper surface **347bUS** of the peripheral separation pattern **347b** may include a second trench **347t**. The etch stop layer **EL** may conformally cover an inner side surface and a lower surface of the second trench **347t**. The third interlayer insulating layer **IL** may include a portion penetrating (or inserted into) the second trench **347t**.

(33) A third width **W3** of the second bit line spacer **336** may be less than a fourth width **W4** of the first bit line spacer **335**. Accordingly, a distance between the second bit line spacers **336** may be increased near an upper portion of the bit line capping pattern **BLC**. Thus, in a method of fabricating a semiconductor device consistent with an embodiment of the inventive concept, the landing pad **LP** may be easily connected to the storage node contact **BC**.

(34) Referring to FIGS. **2A**, **4A**, and **4B**, the landing pad separation pattern **347a** may include a stacked combination of first, second and third (hereafter collectively, "first to third") cell sub-insulating patterns **SBL1a** to **SBL3a**. The peripheral separation pattern **347b** may include a stacked combination of first to third peripheral sub-insulating patterns **SBL1b** to **SBL3b**. The first cell sub-insulating pattern **SBL1a** may include the same material as the first peripheral sub-insulating pattern **SBL1b** and may have a seventh thickness **T7** that is constant. The second cell sub-insulating pattern **SBL2a** may include the same material as the second peripheral sub-insulating pattern **SBL2b** and may have an eighth thickness **T8** that is constant. The third cell sub-insulating pattern **SBL3a** may include the same material as the third peripheral sub-insulating pattern **SBL3b**. The third cell sub-insulating pattern **SBL3a** may fill a space between the landing pads **LP**. The third peripheral sub-insulating pattern **SBL3b** may have a ninth thickness **T9** that is constant. The second cell sub-insulating pattern **SBL2a** and the second peripheral sub-insulating pattern **SBL2b** may include a material which has an etch selectivity with respect to the first and third cell sub-insulating patterns **SBL1a** and **SBL3a** and the first and third peripheral sub-insulating patterns **SBL1b** and **SBL3b**. For example, the second cell sub-insulating pattern **SBL2a** and the second peripheral sub-insulating pattern **SBL2b** may include silicon carbon nitride (**SiCN**). The first and third cell sub-insulating patterns **SBL1a** and **SBL3a** and the first and third peripheral sub-insulating patterns **SBL1b** and **SBL3b** may include silicon nitride. The third peripheral sub-insulating pattern **SBL3b** may include an upper surface, on which the second trench **347t** is formed. In the illustrated example of FIG. **4B**, the sum of the seventh, eighth and ninth thicknesses **T7**, **T8** and **T9** may correspond to the sixth thickness **T6** in the illustrated example of FIG. **3B**. Except for the afore-described differences, the embodiments of **4A** and **4B** may be understood as being substantially the same as the embodiments previously described in relation to FIGS. **3A**, **3B**, **3C** and **3D**.

(35) In some embodiments, the landing pad separation pattern **347a** may be configured to have only the third cell sub-insulating pattern **SBL3a**, thereby omitting the first and second cell sub-insulating patterns **SBL1a** and **SBL2a**.

(36) FIG. **3D** illustrates an embodiment that may be realized by combining the embodiments of FIGS. **4B** and **3C**. Referring to FIG. **3D**, the third peripheral sub-insulating pattern **SBL3b** may have a ninth side thickness **T9s** on the side surface of the peripheral conductive pattern **343bw** and a ninth bottom thickness **T9b** on the lower surface of the first trench **331t**. The ninth bottom thickness **T9b** may be greater than the ninth side thickness **T9s**.

(37) Here, in some embodiments, the peripheral separation pattern **347b** may have a constant sixth thickness **T6**, or a variable sixth thickness that increases (e.g., **T6b** of FIG. **3C**). Accordingly, it may

be possible to prevent an etchant from entering the second interlayer insulating layer **329** during a process of removing a mold layer and to prevent a short circuit issue, thereby improving the reliability of the resulting semiconductor device.

(38) FIG. 2B is a cross-sectional view further illustrating the semiconductor device of FIG. 1 taken along the lines R1-R1' and R2-R2'.

(39) Referring to FIG. 2B, the semiconductor device may include a protection pattern **349p** filling the second trench **347t**. The protection pattern **349p** may include at least one of an insulating material, a semiconductor material, and a conductive material having an etch selectivity with respect to the peripheral separation pattern **347b**. Assuming that the peripheral separation pattern **347b** has the same structure as shown in FIG. 4B, the protection pattern **349p** may contact with the third peripheral sub-insulating pattern **SBL3b**. However, assuming that the peripheral separation pattern **347b** has the same structure as shown in FIG. 3C, the protection pattern **349p** may fill the peripheral recess region **347r**.

(40) The protection pattern **349p** may include at least one of spin-on-hardmask (SOH) material, silicon oxide, silicon oxynitride, silicon germanium, polysilicon, and metal. The protection pattern **349p** may be formed of a material having a dielectric constant less than a dielectric constant of the peripheral separation pattern **347b**. In the case where the protection pattern **349p** is formed of a material having a dielectric constant lower than the dielectric constant of the peripheral separation pattern **347b**, a parasitic capacitance between the peripheral conductive patterns **343bw** may be reduced. In the case where the protection pattern **349p** contains a conductive material, the protection pattern **349p** may serve as a shielding element between the peripheral conductive patterns **343bw**.

(41) In some embodiments, an upper surface of the protection pattern **349p** may be coplanar with the upper surface of the peripheral separation pattern **347b** and the upper surface of the peripheral conductive pattern **343bw**. The etch stop layer EL may cover the protection pattern **349p**. In some embodiments, the etch stop layer EL and the third interlayer insulating layer IL may not penetrate into a region between the peripheral conductive patterns **343bw**. Except for the afore-described differences, the semiconductor device of FIG. 2B may include substantially the same features as those previously described in relation to FIGS. 1, 2A, 3A, 3B, 3C, 3D, 4A and 4B.

(42) FIGS. 5A to 5Q are related cross-sectional views illustrating a method of fabricating a semiconductor device according to embodiments of the inventive concept.

(43) Referring to FIGS. 1 and 5A, the semiconductor substrate **301** including the cell array region CA and the peripheral region PA may be prepared. The device isolation layer **303** may be formed in the semiconductor substrate **301** to define active portions ACTC and ACTP. Although not shown in FIG. 5A, word lines WL (e.g., of FIG. 1) may be formed on the cell array region CA to cross the active regions. In some embodiments, the word lines WL may be buried in the semiconductor substrate **301**. A word line insulating layer may be interposed between the word lines WL and the semiconductor substrate **301**. Word line capping patterns may be disposed on the word lines WL. The first impurity region **305a** and the second impurity region **305b** may be formed in the cell array region CA (e.g., in the semiconductor substrate **301** between the word lines).

(44) The first interlayer insulating layer **307** may be formed on the cell array region CA of the semiconductor substrate **301**. The first interlayer insulating layer **307** may include silicon nitride. Here, the first interlayer insulating layer **307** may be formed to expose an upper surface of the peripheral region PA of the semiconductor substrate **301**. The peripheral gate insulating layer **309** may be formed on the upper surface of the peripheral region PA of the semiconductor substrate **301**. The peripheral gate insulating layer **309** may include at least one of silicon nitride and metal oxide. When the peripheral gate insulating layer **309** is formed, the first interlayer insulating layer **307** may serve as a mask covering the cell array region CA.

(45) Referring to FIG. 5B, a first poly-silicon layer may be deposited to cover the semiconductor substrate **301**. The first poly-silicon layer may be doped with impurities. The first poly-silicon layer

may be patterned to form a first polysilicon mask pattern **311**, which has an opening that defines a position of the cell array region CA, on which a bit line contact DC will be formed. The first polysilicon mask pattern **311** may cover the peripheral region PA. A recess region **313** exposing the first impurity region **305a** may be formed by patterning the first interlayer insulating layer **307** and the semiconductor substrate **301** using the first polysilicon mask pattern **311** as an etch mask. A portion of the device isolation layer **303** may also be etched during this process.

(46) Referring to FIG. 5C, a second poly-silicon layer may be deposited on the semiconductor substrate **301** to fill the recess region **313**. Then, a polishing process may be performed to expose the first polysilicon mask pattern **311** and to form a second polysilicon pattern **314** in the recess region **313**. The second polysilicon pattern **314** may be formed to have an upper surface that is coplanar with an upper surface of the first polysilicon mask pattern **311**. An ohmic layer **315**, a metal-containing layer **317**, and a first capping layer **319** may be sequentially stacked on the second polysilicon pattern **314** and the first polysilicon mask pattern **311**. The ohmic layer **315** may include a metal silicide (e.g., cobalt silicide). The metal-containing layer **317** may include at least one titanium nitride and tungsten. The first capping layer **319** may include silicon nitride. A first mask pattern **321** may be formed on the first capping layer **319**. The first mask pattern **321** may define a peripheral gate electrode on the peripheral region PA. The first mask pattern **321** may be formed to cover the entirety of the cell array region CA. The first mask pattern **321** may include at least one of a photoresist material and a carbon-containing material.

(47) Referring to FIG. 5D, the first capping layer **319**, the metal-containing layer **317**, the ohmic layer **315**, the first polysilicon mask pattern **311**, and the peripheral gate insulating layer **309** on the peripheral region PA may be sequentially patterned using the first mask pattern **321** as an etch mask to expose the upper surface of the semiconductor substrate **301** and to form a peripheral gate insulating pattern **309b**, a peripheral gate electrode **323b**, and a peripheral gate capping pattern **319b**. The peripheral gate electrode **323b** may include a peripheral polysilicon pattern **311b**, a peripheral ohmic pattern **315b**, and a peripheral metal-containing pattern **317b**, which are sequentially stacked.

(48) After the etching process forming the peripheral gate electrode **323b**, the first mask pattern **321** may be removed. Thereafter, a peripheral spacer **325** may be formed to cover side surfaces of the peripheral gate capping pattern **319b**, the peripheral gate electrode **323b**, and the peripheral gate insulating pattern **309b**.

(49) An ion implantation process may be performed to form peripheral source/drain regions **327** in portions of the semiconductor substrate **301** around the peripheral gate electrode **323b**. After the formation of the peripheral source/drain regions **327**, a second interlayer insulating layer **329** may be formed on the semiconductor substrate **301**. The second interlayer insulating layer **329** may include at least one of silicon oxide, silicon oxynitride, silicon nitride, and porous oxide. Thereafter, a polishing process may be performed on the second interlayer insulating layer **329** to expose an upper surface of the first capping layer **319** on the cell array region CA.

(50) Referring to FIGS. 5D and 5E, a second capping layer **331** may be formed to cover the entirety of the semiconductor substrate **301**. The second capping layer **331** may include silicon nitride. A second mask pattern **333** may be formed on the second capping layer **331** to define a region, on which a bit line BL will be formed, on the cell array region CA. The second mask pattern **333** may include at least one of a photoresist material and a carbon-containing material. The second mask pattern **333** may be formed to cover the entirety of the peripheral region PA. The second capping layer **331**, the first capping layer **319**, the metal-containing layer **317**, the ohmic layer **315**, the first polysilicon mask pattern **311**, and the second polysilicon pattern **314** on the cell array region CA may be patterned using the second mask pattern **333** as an etch mask to form the bit line contact DC and a bit line BL, a first bit line capping pattern **319a**, and a second bit line capping pattern **331a**, which are disposed on the bit line contact DC.

(51) An inner side surface and a portion of a lower surface of the recess region **313** may be exposed

by the etching process for forming the bit line BL. In addition, an upper surface of the first interlayer insulating layer **307** may be exposed. In the recess region **313**, a side surface of the bit line contact DC may be exposed.

(52) Referring to FIG. 5F, an insulating layer may be deposited and etched to form a gapfill insulating pattern **316** that substantially fills the recess region **313**. Here, the gapfill insulating pattern **316** may include at least one of silicon nitride, silicon oxide, or silicon oxynitride, and may be a single layer structure or a multi-layered structure. The formation of the gapfill insulating pattern **316** may be performed to expose the upper surface of the first interlayer insulating layer **307**. A spacer layer may be conformally formed on the semiconductor substrate **301**. Then, an anisotropic etching process may be performed to form a first bit line spacer **335** covering side surfaces of the bit line BL, the first bit line capping pattern **319a**, and the second bit line capping pattern **331a**. The first bit line spacer **335** may include at least one of silicon nitride, silicon oxide, and silicon oxynitride, and may be a single layer structure or a multi-layered structure. The first bit line capping pattern **319a** and the second bit line capping pattern **331a** may constitute a bit line capping pattern BLC.

(53) Referring to FIG. 5G, the first interlayer insulating layer **307** between the bit lines BL may be etched to form a storage node contact hole exposing the second impurity region **305b**. During this process, the gapfill insulating pattern **316**, the device isolation layer **303**, and the semiconductor substrate **301** may be partially etched. A doped poly-silicon layer may be deposited to cover the entirety of the semiconductor substrate **301** and to fill the storage node contact hole. Then, an anisotropic etching process, in which a mask (e.g., a photoresist pattern) is not used, may be performed to form a storage node contact BC partially filling the storage node contact hole. The first bit line spacer **335** may be partially removed to expose an upper side surface of the bit line capping pattern BLC. An additional spacer layer may be conformally formed, and then, an etch-back process may be performed to form a second bit line spacer **336** on the first bit line spacer **335**. Here, the second bit line spacer **336** may have a width less than a width of the first bit line spacer **335**.

(54) Referring to FIG. 5H, the second capping layer **331** and the second interlayer insulating layer **329** on the peripheral region PA may be sequentially patterned to form a peripheral contact hole **339** exposing the peripheral source/drain region **327**. A silicidation process may be performed to form a storage node ohmic layer **341a** on the storage node contact BC and to form a peripheral ohmic layer **341b** on an upper surface of the semiconductor substrate **301** exposed by the peripheral contact hole **339**.

(55) Referring to FIG. 5I, a second metal-containing layer **343** may be formed to cover the semiconductor substrate **301** to fill the storage node contact hole and the peripheral contact hole **339**. In some embodiments, the second metal-containing layer **343** may include titanium nitride and tungsten. A mask pattern MK may be formed on the second metal-containing layer **343**. The mask pattern MK may define a shape of a landing pad LP and a shape of a peripheral conductive pattern **343bw**. The mask pattern MK may include at least one of a photoresist, a silicon oxide, a polysilicon, a silicon nitride, a silicon oxynitride, SOH, and an amorphous carbon layer (ACL).

(56) Referring to FIGS. 5I and 5J, the second metal-containing layer **343** may be etched using the mask pattern MK as an etch mask to form the landing pad LP. Here, the landing pad LP is electrically connected to the storage node contact BC in the cell array region CA. During this process, a peripheral conductive pattern **343bw** and a first peripheral contact **343bc** in the peripheral contact hole **339** may be formed on the peripheral region PA. The peripheral conductive pattern **343bw** may be an interconnection line or a contact pad, which is provided in the peripheral circuit region. An upper portion of the second capping layer **331** may be partially removed by the etching process. In this case, a first trench **331t** may be formed in the second capping layer **331**. In addition, the second bit line capping pattern **331a** and the second bit line spacer **336** may also be partially etched by the etching process. Thus, empty spaces GAP may be formed between the

landing pads LP. The empty space GAP may have a first width W1, as illustrated in FIG. 3A. The first trench 331t may have a second width W2 greater than the first width W1, as illustrated in FIG. 3B.

(57) Referring to FIGS. 5J and 5K, a separation insulating layer 347 may be conformally formed to cover the semiconductor substrate 301. The separation insulating layer 347 may be formed with a thickness large enough to fill the empty spaces GAP between the landing pads LP, yet thin enough to fill a space between the peripheral conductive patterns 343bw. The separation insulating layer 347 may be formed to have a constant thickness (e.g., the sixth thickness T6 of FIG. 3B). In some embodiments, like the one illustrated in FIG. 3C, the separation insulating layer 347 may be formed such that the thickness on the lower surface of the first trench 331t (e.g., the sixth bottom thickness T6b of FIG. 3C) is greater than the thickness on the side surface of the peripheral conductive patterns 343bw (e.g., the sixth side thickness T6s of FIG. 3C).

(58) The separation insulating layer 347 may include at least one of silicon nitride and silicon carbon nitride, and may be a single layer structure or a multi-layered structure. A protection layer 349 may be formed on the separation insulating layer 347. The protection layer 349 may include a material having an etch selectivity with respect to the separation insulating layer 347. For example, the protection layer 349 may be formed of SOH materials having excellent gap-filling properties. Alternately, the protection layer 349 may include at least one of silicon oxide, silicon oxynitride, silicon germanium, polysilicon, and a metal-containing material. The protection layer 349 may be formed to include a portion penetrating into a region between the peripheral conductive patterns 343bw.

(59) Referring to FIGS. 5K and 5L, an unmasked etch-back process or a chemical mechanical polishing (CMP) process may be performed on the protection layer 349 to remove the protection layer 349 on the separation insulating layer 347, to expose the upper surface of the separation insulating layer 347, and to form a protection pattern 349p between the peripheral conductive patterns 343bw. Thereafter, an etching process (e.g., an unmasked etch-back process) may be performed on the separation insulating layer 347 to remove a portion of the separation insulating layer 347, to expose the upper surfaces of the landing pads LP and the upper surfaces of the peripheral conductive patterns 343bw, to form a landing pad separation pattern 347a between the landing pads LP, and to form a peripheral separation pattern 347b between the peripheral conductive patterns 343bw. Here, the protection pattern 349p may prevent the peripheral separation pattern 347p under the protection pattern 349p from being partially etched during the etching process.

(60) Referring to FIGS. 5L and 5M, the peripheral separation pattern 347b may be exposed by selectively removing only the protection pattern 349p. An etch stop layer EL, a mold layer ML, and a supporting layer SL may be sequentially stacked on the semiconductor substrate 301. The etch stop layer EL may include silicon nitride. The mold layer ML may include silicon oxide. The supporting layer SL may include silicon nitride.

(61) Referring to FIG. 5N, the supporting layer SL, the mold layer ML, and the etch stop layer EL on the cell array region CA may be etched to form bottom electrode holes BEH exposing the landing pads LP. A conductive layer may be formed to cover the entirety of the semiconductor substrate 301. Then, a CMP process or an etch-back process may be performed to form bottom electrodes BE in the bottom electrode holes BEH. The bottom electrodes BE may be disposed to have island-shaped structures that are spaced apart from each other, when viewed in a plan view. Each of the bottom electrodes BE may be shaped like a pillar, a plug, or a hollow cylinder.

(62) Referring to FIGS. 5N and 5O, the supporting layer SL may be etched to expose a portion of an upper surface of the mold layer ML and to form a supporting pattern SP. The supporting pattern SP may be in partial contact with upper side surfaces of the bottom electrodes BE. The supporting layer SL may be completely removed from the peripheral region PA.

(63) Referring to FIGS. 3B, 5O, and 5P, an isotropic etching process may be performed to remove

the mold layer ML and to expose upper and side surfaces of the bottom electrodes BE, upper and lower surfaces of the supporting pattern SP, and an upper surface of the etch stop layer EL. In some embodiments, the isotropic etching process may be performed in an over-etching manner to completely remove the mold layer ML. In some regions, not only a portion of the etch stop layer EL, but also a portion of the peripheral separation pattern **347b** may be removed by an etchant used during the isotropic etching process.

(64) If the anisotropic etching process is performed on the separation insulating layer **347** without forming the protection pattern **349p** (e.g., the method step of FIG. 5L), a peripheral separation pattern **347b** may be formed having a spacer shape covering the side surfaces of the peripheral conductive patterns **343bw**, and the second capping layer **331** of a fourth remaining thickness **T4r** (e.g., see FIG. 3B) may be exposed through the first trench **331t**. In this case, not only a portion of the etch stop layer EL, but also the exposed portion of the second capping layer **331** under the first trench **331t** may be removed by an etchant used for over-etching of the mold layer ML.

Furthermore, the second interlayer insulating layer **329** may be etched by the etchant, which is used to remove the mold layer ML. In this case, empty spaces may be formed in the second interlayer insulating layer **329**. When a conductive layer is deposited in a subsequent step, the empty spaces may be filled with the conductive layer, thereby creating an unwanted conductive path, and causing a short circuit.

(65) In great contrast, embodiments of the inventive concept include the formation of the protection pattern **349p**, such that the peripheral separation pattern **347b** may be formed with a position-independent, constant thickness **T6** (or alternately an increasing vertical thickness **T6b**—e.g., see FIG. 3C). Thus, it may be possible to prevent the etchant used to remove the mold layer ML from coming into contact with the second interlayer insulating layer **329**. Accordingly, it may be possible to reduce the possibility of creating a process failure, thereby increasing manufacturing yield.

(66) Referring to FIG. 5Q, a dielectric layer DL may be formed on the cell array region CA to conformally cover the bottom electrodes BE and the supporting pattern SP. Then, an upper electrode UE may be formed on the dielectric layer DL. Next, a third interlayer insulating layer IL may be formed to cover the entirety of the semiconductor substrate **301**. Thereafter, a cell contact CT1, a cell interconnection line UM1, a second peripheral contact MC, and a peripheral interconnection line UM2 may be formed, as shown in FIG. 2A.

(67) The semiconductor device of FIG. 2B may be formed by performing a subsequent processes (e.g., of forming the etch stop layer EL) while the protection pattern **349p** is not removed in the step of FIG. 5L.

(68) In methods of fabricating embodiments of the inventive concept like the semiconductor devices of FIGS. 4A and 4B, the step of forming the separation insulating layer **347** (e.g., of FIG. 5K) may include conformally and sequentially stacking first to third sub-insulating layers. In the step of FIG. 5L, the etch-back process on the separation insulating layer **347** may include performing the etch-back process on the first to third sub-insulating layers to form the first to third cell sub-insulating patterns SBL1a to SBL3a and the first to third peripheral sub-insulating patterns SBL1b to SBL3b of FIGS. 4A and 4B. Except for these differences, other steps and features of the fabrication methods according to the embodiments of the inventive concept may be performed in substantially the same or similar manner(s).

(69) The first cell and peripheral sub-insulating patterns SBL1a and SBL1b may be formed as a result of the etching of the first sub-insulating layer. The second cell and peripheral sub-insulating patterns SBL2a and SBL2b may be formed as a result of the etching of the second sub-insulating layer. The third cell and peripheral sub-insulating patterns SBL3a and SBL3b may be formed as a result of the etching of the third sub-insulating layer. The second sub-insulating layer may include a material having an etch selectivity with respect to the first and third sub-insulating layers. For example, the second sub-insulating layer may be formed of silicon carbon nitride (SiCN), and the first and third sub-insulating layers may be formed of silicon nitride. The second sub-insulating

layer may serve as an etch stop layer in the etch-back process on the separation insulating layer 347.

(70) According to embodiments of the inventive concept, a semiconductor device may include a peripheral separation pattern with a thickness that is constant regardless of position, therefore the reliability of the semiconductor devices is improved. In a method of fabricating semiconductor devices according to embodiments of the inventive concept, by forming a protection pattern, it is possible to form a peripheral separation pattern having a constant thickness regardless of position (or alternately an increasing vertical thickness). Accordingly, it is possible to prevent an etchant used during the removing of the mold layer from entering an underlying insulating layer, thereby reducing the possibility of creating a process failure. As a result, manufacturing yield improves.

(71) In the foregoing written description, some material layers, elements and/or components have been described as “including” one or more of a variety of suggested material(s)—e.g., metal(s), metal silicide(s), polysilicon(s), oxide(s), nitride(s), doped-polysilicon(s), etc. Those skilled in the art will appreciate that these examples are merely illustrative in nature, and are intended to teach or suggest a variety of options that practitioners seeking to make embodiments of the inventive concept may use. In some instances, the term “include” (or “including”) denotes a “forming from” one or more of the suggested materials(s).

(72) In the foregoing written description, some material layers, elements and/or components may be described as being “equal” in some physical attribute or characteristic—e.g., height, level, width, length, thickness, size, etc. Those skilled in the art will recognize that the term “equal” in this regard means “substantially equal”—allowing for routine variations in the fabrication of material layers, elements and/or components. In like manner, certain material layers, elements and/or components may be described as including the “same material.” Those skilled in the art will recognize that the term “same material” in this regard means “substantially the same material”—allowing for routine variations in the composition of material layers, elements and/or components.

(73) While embodiments of the inventive concept have been particularly shown and described, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and scope of the inventive concept as defined by the following claims.

Claims

1. A semiconductor device, comprising: a substrate including source/drain region; a gate electrode on the substrate; an interlayer insulating layer covering the gate electrode on the substrate; a plurality of conductive contacts penetrating the interlayer insulating layer, each conductive contact of the plurality of conductive contacts being electrically connected to the source/drain region; a plurality of conductive lines on the plurality of conductive contacts; a separation pattern structure interposed between the plurality of conductive lines on the gate electrode; and an etch stop layer covering the plurality of conductive lines and the separation pattern structure, wherein the plurality of conductive contacts and the plurality of conductive lines include the same materials, wherein the separation pattern structure includes: a first separation pattern having a trench and a second separation pattern filling the trench, wherein the first separation pattern and the second separation pattern include different materials, and wherein a width of the separation pattern structure in a first direction parallel to a top surface of the substrate is equal to or smaller than a width of the gate electrode in the first direction.
2. The semiconductor device of claim 1, wherein a dielectric constant of the second separation pattern is lower than a dielectric constant of the first separation pattern.
3. The semiconductor device of claim 1, wherein: the first separation pattern includes at least one of silicon nitride and silicon carbon nitride and the second separation pattern includes at least one of spin-on-hardmask (SOH) materials, silicon oxide, silicon oxynitride, silicon germanium,

polysilicon or metal-containing materials.

4. The semiconductor device of claim 1, further comprising a capping layer on the interlayer insulating layer, wherein the first separation pattern penetrates a portion of the capping layer.

5. The semiconductor device of claim 1, further comprising an ohmic pattern between the plurality of conductive contacts and the source/drain region.

6. The semiconductor device of claim 1, wherein the second separation pattern disposed to be spaced apart from the plurality of conductive lines.

7. The semiconductor device of claim 1, wherein a level of an upper surface of the separation pattern structure is substantially same as a level of an upper surface of the plurality of conductive lines.

8. The semiconductor device of claim 1, wherein: the plurality of conductive contacts and the plurality of conductive lines include at least one of titanium nitride and tungsten.

9. A semiconductor device, comprising: a substrate including a cell array region and a peripheral region; a gate electrode on the peripheral region; a first interlayer insulating layer covering the gate electrode on the peripheral region; a capping layer on the first interlayer insulating layer; a plurality of conductive contacts penetrating the first interlayer insulating layer and the capping layer; a plurality of conductive lines spaced apart from each other on the capping layer and the plurality of conductive contacts; a separation pattern structure interposed between the plurality of conductive lines on the first interlayer insulating layer; an etch stop layer covering the plurality of conductive lines and the separation pattern structure; and a second interlayer insulating layer on the etch stop layer, wherein the separation pattern structure includes a first separation pattern and a second separation pattern surrounding a lower surface and a side surface of the first separation pattern on the first separation pattern, and wherein the first separation pattern and the second separation pattern include different materials.

10. The semiconductor device of claim 9, further comprising a peripheral contact penetrating the etch stop layer and the second interlayer insulating layer, wherein the peripheral contact is in contact with at least one of the plurality of conductive lines.

11. The semiconductor device of claim 9, wherein an upper surface of the first separation pattern and an upper surface of the second separation pattern are in contact with the etch stop layer.

12. The semiconductor device of claim 9, wherein at least a portion of a side surface of the second separation pattern is in contact with the plurality of conductive lines.

13. The semiconductor device of claim 9, wherein a level of an upper surface of the separation pattern structure is substantially same with a level of an upper surface of the plurality of conductive lines.

14. The semiconductor device of claim 9, wherein the plurality of conductive contacts and the plurality of conductive lines include at least one of titanium nitride and tungsten.

15. The semiconductor device of claim 9, wherein: the first separation pattern includes spin-on-hardmask (SOH) materials, silicon oxide, silicon oxynitride, silicon germanium, polysilicon or metal-containing materials and the second separation pattern includes at least one of silicon nitride and silicon carbon nitride.

16. The semiconductor device of claim 9, wherein the gate electrode and the separation pattern structure vertically overlap.
